Lecture L05 — score on row L05 in the Excel form

Normalization Forms and Database Schema Design (1NF–BCNF)

Learning Objectives:

- Understand the definition of normalization forms.
- Apply the concept of 1NF to a database schema.
- Explain the concept of 2NF.
- Apply the concept of 2NF to a database schema.
- Understand the concept of 3NF.
- Apply the concept of 3NF to a database schema.
- Explain the concept of BCNF.
- Apply the concept of BCNF to a database schema.

Academic Accuracy

Database normalization is a process of organizing data into a relational database to minimize data redundancy and dependency. The process involves breaking down large tables into smaller, more manageable tables. This helps in maintaining data integrity and minimizing the risk of data anomalies.

1. Normal Form (1NF)

1NF ensures that every column in a table contains atomic values, meaning that each value in a column is a single piece of data and cannot be divided into multiple pieces. An example of a table in 1NF is:

|c|c|

Employee ID | Name

1 | John Doe

2 | Jane Doe

In this table, each column contains a single atomic value.

2. Second Normal Form (2NF)

2NF requires that every non-key attribute (attribute not in the primary key) depends solely on the primary key. In other words, all non-key attributes must be fully functionally dependent on the primary key. An example of a table in 2NF is:

|c|c|c|

Employee ID | Name | Age

1 | John Doe | 30

2 | Jane Doe | 35

In this table, "Age" is a non-key attribute, and it depends solely on the primary key (Employee ID).

3. Third Normal Form (3NF)

3NF ensures that a table in 2NF is in a transitive dependency, meaning that for every non-key attribute, there is a direct dependency on the primary key. An example of a table in 3NF is:

|c|c|c|

Employee ID | Name | Age | Department

1 | John Doe | 30 | IT

2 | Jane Doe | 35 | HR

In this table, "Age" is a non-key attribute, and it depends directly on the primary key (Employee ID) through the "Department" column.

4. Boyce-Codd Normal Form (BCNF)

BCNF is the highest form of normalization and ensures that every non-key attribute is fully functionally dependent on the primary key. It is achieved by removing multi-valued dependencies (MVDs) and ensuring that all non-key attributes are fully functionally dependent on the primary key. An example of a table in BCNF is:

|c|c|

Employee ID | Name

1 | John Doe

2 | Jane Doe

In this table, "Name" is a non-key attribute, and it depends directly on the primary key (Employee ID).

5. Functional Dependency (FD)

A functional dependency occurs when a non-key attribute (attribute not in the primary key) is dependent on a key attribute (attribute in the primary key). It is represented by a symbol (←) between the non-key attribute and the key attribute.

6. Multivalued Dependency (MVD)

An MVD occurs when a non-key attribute (attribute not in the primary key) depends on more than one key attribute (attributes in the primary key). It is represented by multiple symbols (←) between the non-key attribute and the key attributes.

7. Key (Key Attribute)

A key is a combination of attributes in a table that uniquely identifies each record in the table. It is called the primary key if it uniquely identifies each record, and it is called a foreign key if it is used to reference records in another table.

Practical Relevance

The concept of normalization forms is crucial in designing database schema. It helps in minimizing data redundancy and dependency, which enhances data integrity and reduces the risk of data anomalies. By applying the principles of normalization forms, database designers can create efficient and reliable database schema that can handle large volumes of data.

1. Normal Form (1NF)

1NF is essential for ensuring that every column contains atomic values, meaning that each value in a column is a single piece of data and cannot be divided into multiple pieces. This ensures data consistency and minimizes the risk of data anomalies.

2. Second Normal Form (2NF)

2NF is important for ensuring that every non-key attribute depends solely on the primary key. This ensures data integrity and reduces the risk of data anomalies.

3. Third Normal Form (3NF)

3NF is essential for ensuring that a table in 2NF is in a transitive dependency, meaning that for every non-key attribute, there is a direct dependency on the primary key. This ensures data consistency and minimizes the risk of data anomalies.

4. Boyce-Codd Normal Form (BCNF)

BCNF is important for ensuring that every non-key attribute is fully functionally dependent on the primary key. This ensures data integrity and reduces the risk of data anomalies.

5. Functional Dependency (FD)

Functional dependency is important for ensuring that a non-key attribute is dependent on a key attribute. This ensures data consistency and minimizes the risk of data anomalies.

6. Multivalued Dependency (MVD)

MVD is important for ensuring that a non-key attribute does not depend on more than one key attribute. This ensures data consistency and minimizes the risk of data anomalies.

7. Key (Key Attribute)

A key is important for ensuring that each record in a table is uniquely identified. This ensures data integrity and reduces the risk of data anomalies.

8. Data Anomalies

Data anomalies are situations where data in a database is inconsistent or incorrect. They can arise due to data redundancy and dependency. By applying the principles of normalization forms, database designers can avoid data anomalies and ensure the accuracy and reliability of data in the database.

Limitations and Common Pitfalls

While normalization forms are essential for ensuring data integrity and minimizing data anomalies, they are not without limitations. Some common pitfalls include:

1. **Over-normalization:** Over-normalization can lead to an excessive number of tables, which can make the database schema more complex and difficult to maintain.

2. **Under-normalization:** Under-normalization can lead to data redundancy, which can increase the size of the database and slow down query performance.

3. **Misapplication of Normalization Forms:** Misapplication of normalization forms can lead to data inconsistencies and data anomalies.

Limitations

The lecture lacks detailed explanations of the limitations and variants of each normal form. The limitations and variants of each normal form are not covered.

Variants of Each Normal Form

1. **Normalization Forms Variants:**

- **1NF Variants:**
- Minimal 1NF: A table in 1NF but with multiple non-key attributes.
- **2NF Variants:**
- Minimal 2NF: A table in 2NF but with multiple non-key attributes.
- **3NF Variants:**
- Minimal 3NF: A table in 3NF but with multiple non-key attributes.
- **BCNF Variants:**
- Minimal BCNF: A table in BCNF but with multiple non-key attributes.

By covering these variants, the lecture can provide a more comprehensive understanding of normalization forms and their application in database schema design.